

CHARLES BENELLO

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EDUCATION

The University of Chicago

Chicago, IL

Master of Science in Financial Mathematics

Expected Mar 2027

- Wayne Booth Graduate Student Teaching Prize (2026) - UChicago's top graduate teaching award
- Coursework: Distributed Systems, ML in SWE, Agentic AI, Reinforcement Learning, Computing for Finance

The University of Chicago | GPA: 3.75/4.0

Chicago, IL

Bachelor of Science in Mathematics and Computer Science, Minor in Korean

Jun 2025

- Coursework: Operating Systems, Computer Architecture, Networks, Database Systems, Algorithms, Concurrency Control

EXPERIENCE

Software Engineering Intern

Feb 2026 – Present

Alter Domus

Chicago, IL

- Architected a multi-stage AI pipeline for 100+ page financial documents with schema-enforced output and cross-reference verification, lifting extraction accuracy from 50% to 92–97%.
- Built an LLM-as-a-Judge evaluation framework with automated benchmarks for faithfulness, numerical accuracy, and completeness; cut LLM inference costs by optimizing context handling.

GPU Systems Researcher

Jun – Sep 2025

EPFL (DIAS Lab, Prof. Anastasia Ailamaki)

Lausanne, Switzerland

- Engineered a macroblock-based GPU decompression strategy in CUDA/C++ on NVIDIA hardware, cutting out-of-memory query times by 50–200% for large analytical workloads.
- Built cost-based simulation models predicting query performance across workload complexities; designed TPC-DS join patterns with resource contention models across I/O, decompression, and compute.

Quantitative Researcher

Mar 2026 – Present

PayPal (UChicago Project Lab)

Chicago, IL

- Built a self-supervised transaction foundation model (TabBERT-style hierarchical 2-level transformer, ~5M params) on 14.5M unlabeled transactions; achieved AUC 0.974 / F1 0.822 on downstream fraud detection.
- Ran 87+ pretraining ablations on UChicago's Midway3 SLURM cluster, automating job submission, log collection, and metrics aggregation across architecture, tokenization, and masking sweeps.

Database Research Assistant

Apr 2023 – Present

ChiData Research Lab, University of Chicago

Chicago, IL

- First-authored VLDB 2026 paper (CrocSort): designed a parallel external merge sort in Rust that completes within memory budgets where PostgreSQL, DuckDB, and ClickHouse abort.
- Co-authored CIDR 2025 paper on resource-adaptive query execution with paged memory management; sole-authored HPTS 2026 paper on cost-based sort configuration for cloud and disaggregated hardware.

Quantitative Researcher

Oct – Dec 2025

Bank of America (UChicago Project Lab)

Chicago, IL

- Shipped full-stack interfaces (Next.js, Flask, React Native, FastAPI) and a multi-agent LLM debate system generating macro stress scenarios for HQLA portfolio rebalancing.

PROJECTS

GPU External Sort | CUDA, C++ | github.com/cmбенello/gpu-research

2025–Present

- Built a GPU external merge sort in CUDA (port of CrocSort, VLDB 2026). On TPC-H SF50 ORDER BY lineitem (300M rows, 36 GB) beats DuckDB by 20.1× and Polars by 34.5× on a single GPU; 2.4× faster than MendSort (best published JouleSort 2023). Cost-based plan-time selector between compact-key and full-key paths; cut PCIe amplification to 0.14× via key-only upload + permutation gather + triple-buffered streaming.

Concurrent Database Engine | Rust | github.com/cmбенello/CrustyDB

2025

- Built a multi-threaded relational database from scratch in Rust: B+ tree indexing, concurrent buffer pool with page-level locking, heap storage, and a SQL query execution engine.

Transit Coverage Optimizer | Python | cmбенello.github.io/covtsp

2026

- Beat the Guinness World Record by 60 minutes (16h45m vs. 17h45m) with an automated Covering TSP pipeline on a time-expanded graph (160K nodes, 1.26M edges). Generalized to NYC and Berlin.

TECHNICAL SKILLS

Languages: Python, C/C++, CUDA, Rust, Go, TypeScript, JavaScript, SQL, Bash

Tools & Infrastructure: Docker, Linux, Git, SLURM, PyTorch, FastAPI, Next.js, React, PostgreSQL, DuckDB

Concepts: GPU Programming, Distributed Systems, AI/ML Infrastructure, Performance Engineering, CI/CD, Cluster Computing, System Administration

PUBLICATIONS

[1] R. Otaki*, C. Benello*, et al. “CrocSort: Resource-Efficient, Skew-Resilient Parallel External Merge Sort.” *VLDB 2026*. (*Equal contribution)

[2] C. Benello. “Getting the Most Out of External Sorting in Cloud and Disaggregated Environments.” *HPTS 2026*.

[3] R. Otaki, J.H. Chang, C. Benello, A. Elmore, G. Graefe. “Resource-Adaptive Query Execution with Paged Memory Management.” *CIDR 2025*.

ADDITIONAL

Languages: Korean, Italian (Intermediate); French (Basic) | **Interests:** Running (Chicago Marathon '22, '23, '24), Climbing, Weightlifting